9 The half-life of the first-order gaseous reaction in which M_2 molecules become converted into M atoms is 40 minutes. 1 mol of M_2 is put into a sealed vessel at pressure *p*.

What will be true when 87.5% of M_2 has been converted into M atoms?

1 80 minutes have elapsed.

2 1.5 mol of M have been formed.

3 The total pressure is $\frac{15}{8}p$ (at constant pressure).

A 1, 2 and 3

B 1 and 2 only

C 2 and 3 only

D 3 only